Swiss Institute of Bioinformatics

SINGLE-CELL TRANSCRIPTOMICS WITH R

Single-cell RNA-seq and Cell Ranger

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Adapted from previous year courses

elixir SWITZERLAND

Why single-cell RNA-seq?

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Single-cell RNA-seq (scRNA-seq) allows us to evaluate the transcriptome at the level of individual cells.

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- To explore which cell types are present in a tissue
- To identify unknown/rare cell types or states
- To elucidate the changes in gene expression during differentiation processes or across time or states
- To identify genes that are differentially expressed in particular cell types
  - between conditions (e.g. treatment or disease)

Bulk vs. Single-cell RNA-seq

- Bulk RNA-seq provides an overview of **average differences in gene expression**. For certain scenarios this has proven to be sufficient (i.e. biomarkers for cancer).

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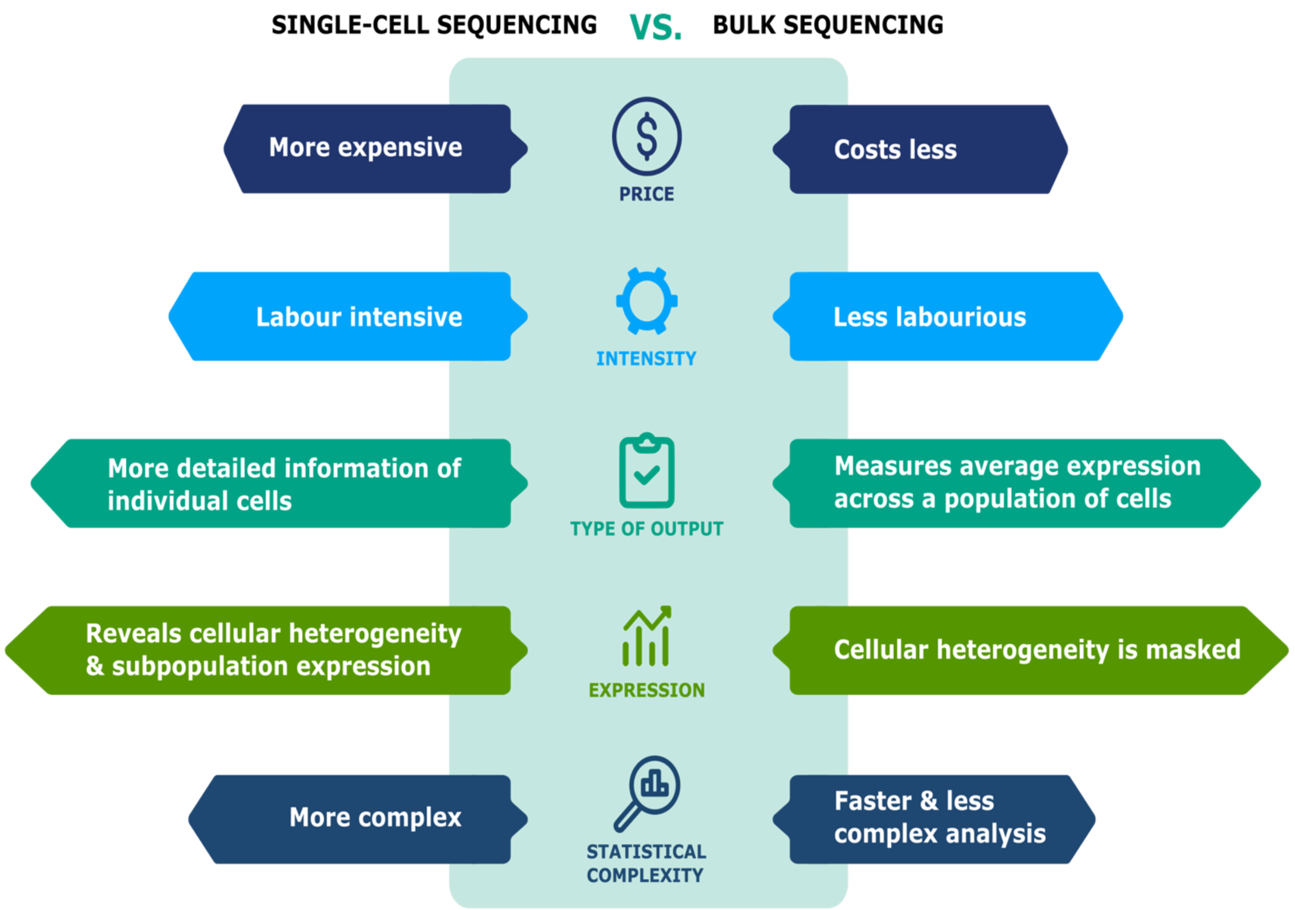

Challenges with scRNA-seq data

- Large volume of data
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Increased complexity and richer datasets, means more room for misinterpretations and deriving wrong conclusions!

Recommendations

- Do not perform single-cell RNA-seq unless it is necessary for the experimental question of interest.

    Could you answer the question using bulk sequencing, which is simpler and less costly? Perhaps FACS sorting the samples could allow for bulk analysis?

Recommendations

- ❖ Understand the details of the experimental question you wish to address.

  Library preparation method and analysis workflow can vary based on the specific experiment and tissue

Recommendations

- Avoid technical sources of variability, if possible:
  - Discuss experimental design with experts prior to the initiation of the experiment
  - Isolate RNA from samples at same time
  - Prepare libraries at same time or alternate sample groups to avoid batch confounding

Experimental design

Replication, randomization and blocking

Be aware of confounding factors, e.g.:

- Person performing handling
- Reagents
- Sequencing lane/library

Record any factor for downstream correction: Taking notes = the best QC

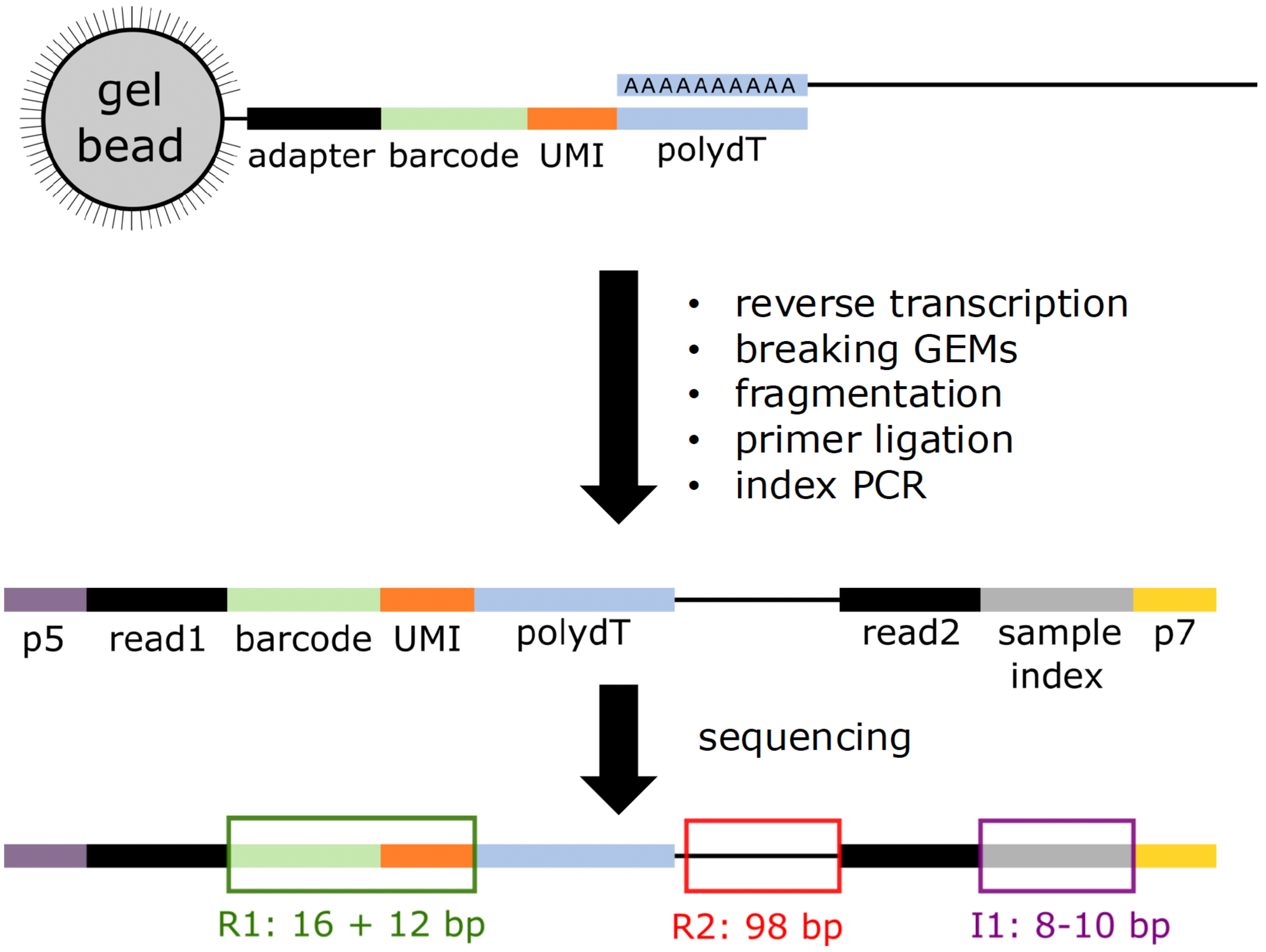
Different cells of one replicate ≠ replicates!

Further reading:

- https://doi.org/10.3389/fcell.2018.00108
- https://doi.org/10.1093/bib/bby007
- https://doi.org/10.1093/bfgp/elx035

Sequencing output

Sequencing output



Sequencing output

sample#   read type

```
ETV6-RUNX1_1_S1_L001_I1_001.fastq.gz
ETV6-RUNX1_1_S1_L001_R1_001.fastq.gz
ETV6-RUNX1_1_S1_L001_R2_001.fastq.gz
```

sample ID   lane

Index Read (I1) - contains cell barcodes + UMI
Read 1 (R1) - contains cell barcode + UMI (same as I1 in 10x v2/v3)
Read 2 (R2) - contains cDNA insert (gene/transcript sequence)

From FASTQ to counts

Tools for this part of the workflow include Alevin, UMI-tools, and Cell Ranger (10X data). While each tool will do things slightly differently the steps below are common to all:

1. Assign reads to cell
2. Alignment
3. Quantification: # UMI/gene → UMI: Unique Molecular Identifier
4. Cell calling

The 10X Software Suite

Pipeline for mapping, filtering, QC and quantitation of libraries

Cell Ranger

Barcode Extraction and filtering

- Identifies cell level barcodes

Mapping to reference

- Uses STAR aligner

Generate count table

- UMIs per gene in each cell

Dimensionality Reduction

- PCA and tSNE

Clustering

- K-means and Graph Based

Cell Ranger

Cell Ranger Count (quantitates a single run)

```
$ cellranger count --id=COURSE \
                   --transcriptome=/cellranger/references/GRCh38/ \
                   --fastqs=/10X/ \
                   --localcores=8 \
                   --localmem=32
```

References:

- Human & mouse: download pre-built from 10x website
- Other organisms: custom reference with **cellranger mkref**

extensive documentation:
https://support.10xgenomics.com/single-cell-gene-expression/software/pipelines/latest/what-is-cell-ranger

Output files generated by Cell Ranger

`web_summary.html` - Web format QC report

`raw/filtered_features_bc_matrix`
- `barcodes.tsv.gz` - cell level barcodes seen in this sample
- `features.tsv.gz` - list of quantitated features (usually Ensembl genes)
- `matrix.mtx.gz` - (sparse) matrix of counts for cells and features

`possorted_genome_bam.bam` - BAM file of mapped reads

`molecule_info.h5` – Details of the cell barcodes

`cloupe.cloupe` - Analysis data for Loupe Cell browser

Cellranger report

ETV6-RUNX1_1

Alerts

The analysis detected ⚠ 1 warning.

	Alert	Value	Detail
⚠	Fraction of RNA read bases with Q-score >= 30 is low	59.4%	Fraction of RNA read bases with Q-score >= 30 should be above 65%. A lower fraction might indicate poor sequencing quality. This is Read 1 for the Single Cell 3' v1 chemistry and Single Cell 5' paired end, Read 2 for the Single Cell 3' v2/v3 chemistry and Single Cell 5' R2-only)

Summary | Analysis

3,091
Estimated Number of Cells

68,259
Mean Reads per Cell

1,717
Median Genes per Cell

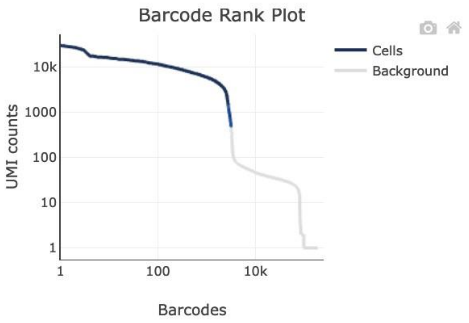
Sequencing

Number of Reads	210,987,037
Number of Short Reads Skipped	0
Valid Barcodes	98.2%
Valid UMIs	100.0%
Sequencing Saturation	84.4%
Q30 Bases in Barcode	96.4%
Q30 Bases in RNA Read	59.4%
Q30 Bases in UMI	96.5%

Mapping

Reads Mapped to Genome	95.8%
Reads Mapped Confidently to Genome	92.9%
Reads Mapped Confidently to Intergenic Regions	5.2%
Reads Mapped Confidently to Intronic Regions	25.5%
Reads Mapped Confidently to Exonic Regions	62.2%
Reads Mapped Confidently to Transcriptome	58.2%
Reads Mapped Antisense to Gene	1.2%

Cells

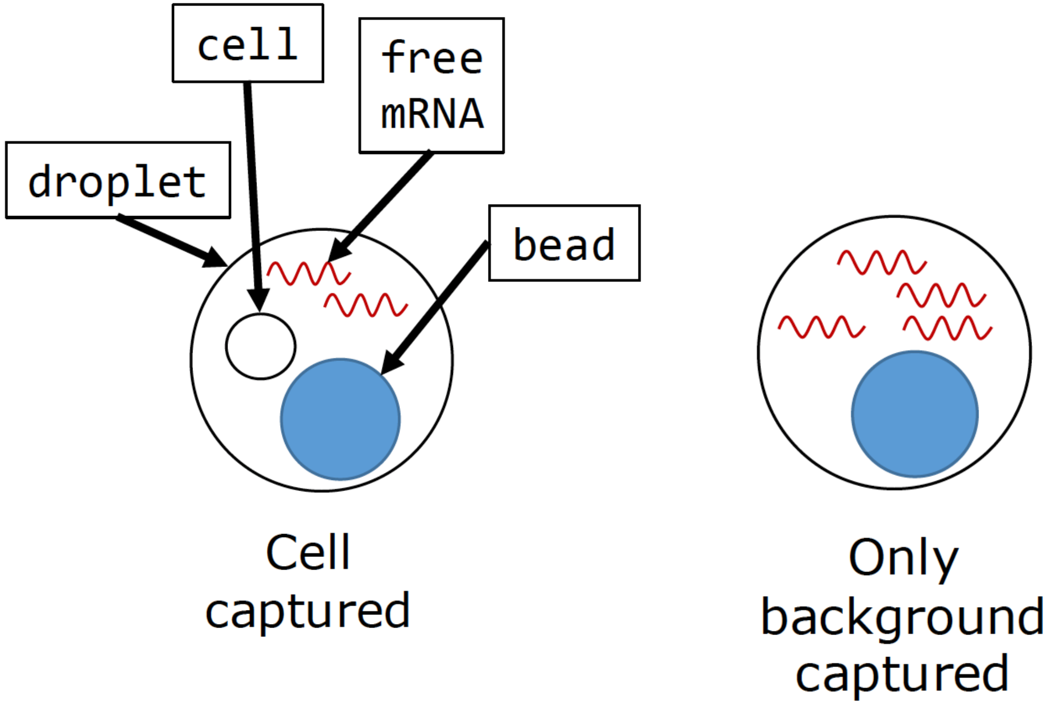


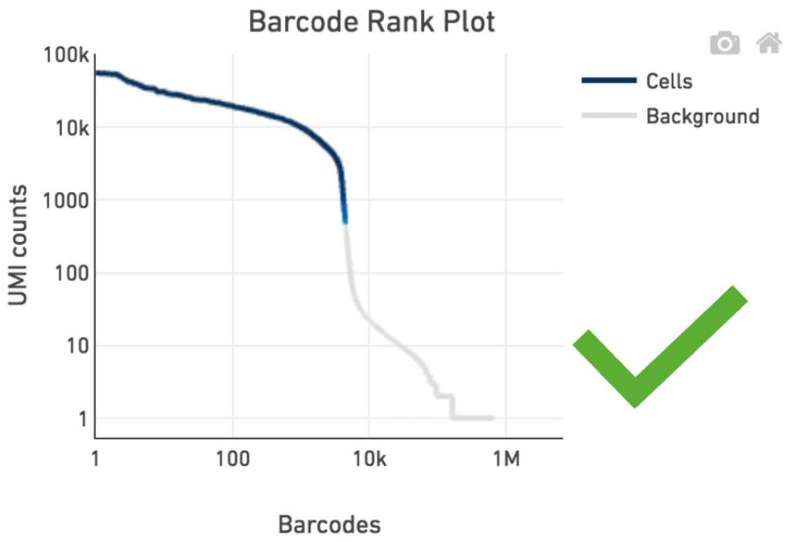
Estimated Number of Cells	3,091
Fraction Reads in Cells	91.1%
Mean Reads per Cell	68,259
Median Genes per Cell	1,717
Total Genes Detected	18,334
Median UMI Counts per Cell	4,825

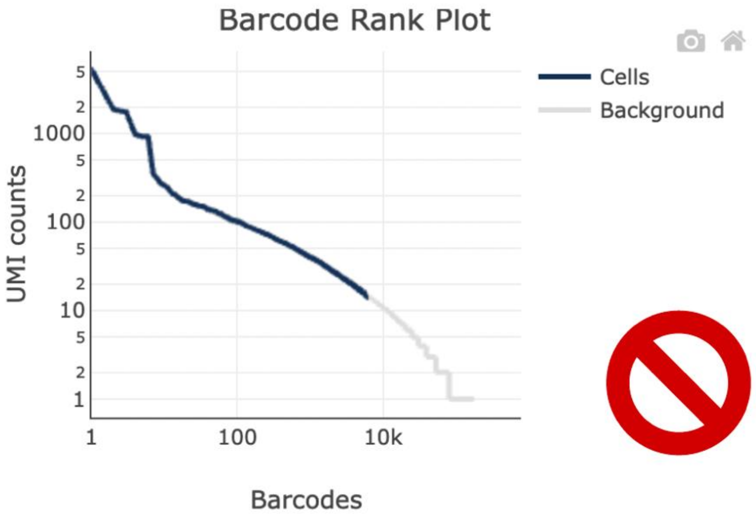
Sample

Sample ID	ETV6-RUNX1_1
Sample Description	
Chemistry	Single Cell 3' v2
Include introns	False
Reference Path	…nger/refdata-cellranger-GRCh38-3.0.0
Transcriptome	GRCh38-3.0.0
Pipeline Version	cellranger-6.0.1

Cell calling







Background 'cells': low #UMI/barcode

Other important parameters

- Number of cells (<8% doublet rate):

  <20k/channel (chromium X)

  <10k/channel (chromium controller)

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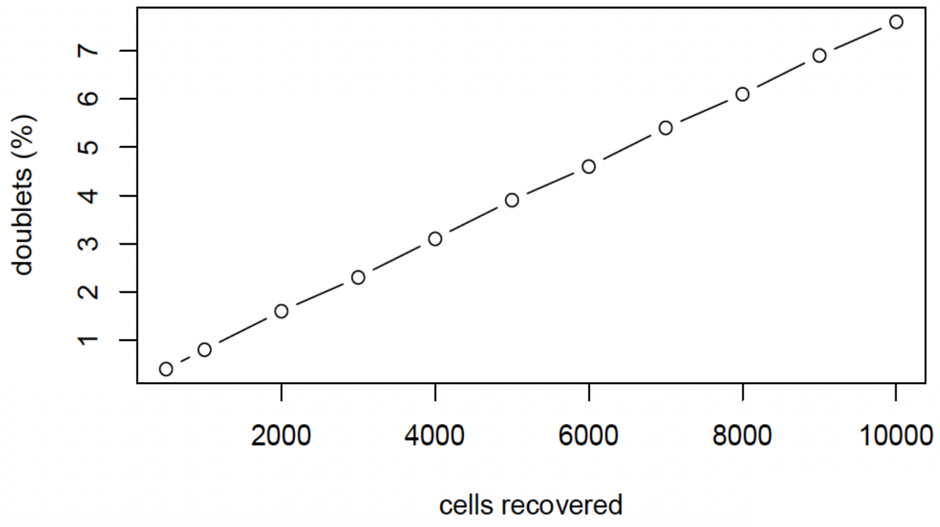
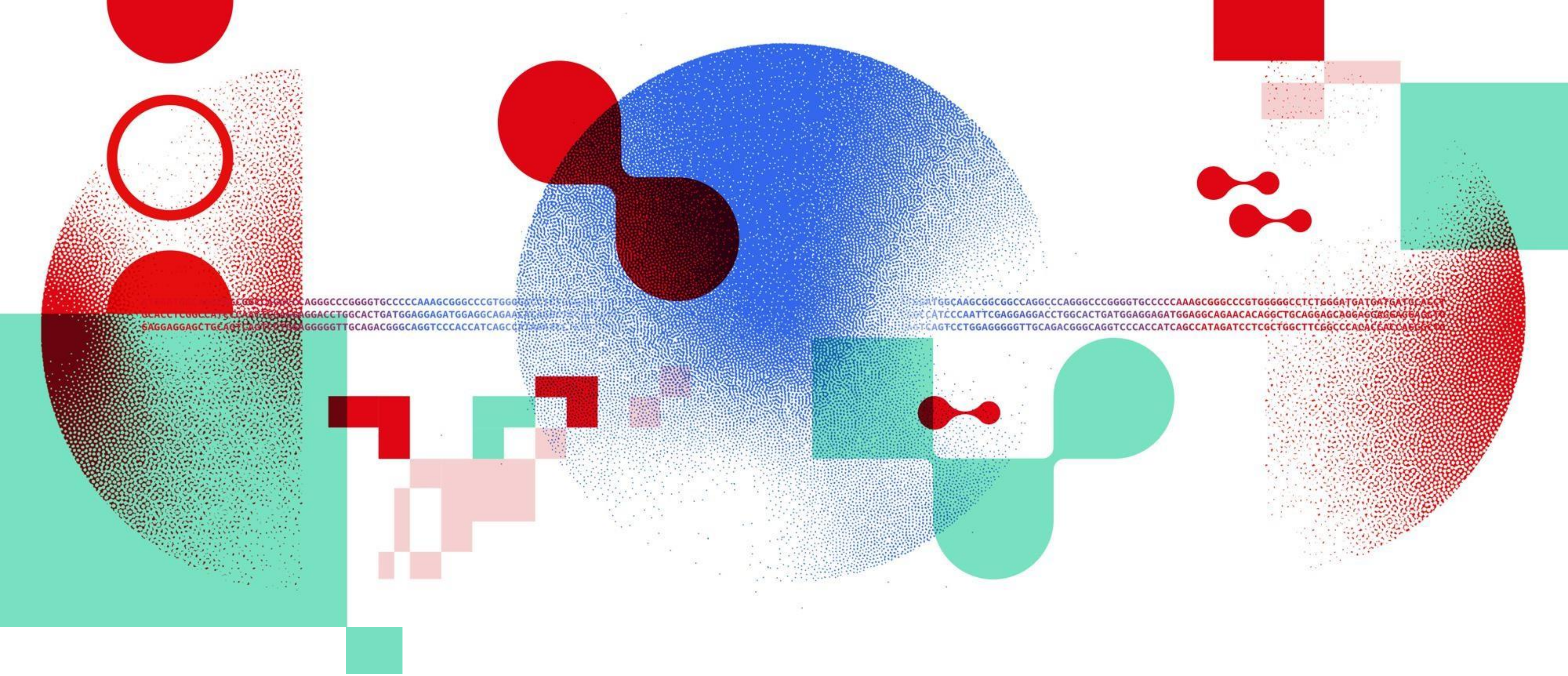
- Sequencing saturation

$$saturation = 1 - \frac{\#\, unique\ reads}{\#\, reads}$$





Thank you

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